

BAYAN-OVOO, Mongolia

WMO: 443020

Lat: 47.7858N

Lon: 112.1106E

Elev: 926

StdP: 90.68

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.9	(c) -30.7	(d) -35.8	(e) 0.1	(f) -32.6	(g) -33.6	(h) 0.2	(i) -30.3	(j) 12.3	(k) -16.7	(l) 11.1	(m) -14.2	(n) 2.7	(o) 270	(p) 0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.2	(c) 32.2	(d) 17.2	(e) 29.9	(f) 16.5	(g) 27.9	(h) 15.9	(i) 19.3	(j) 27.6	(k) 18.4	(l) 26.0	(m) 17.5	(n) 24.6	(o) 4.0	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.9	13.5	21.7	15.9	12.7	20.7	15.0	11.9	20.4	59.3	27.8	56.1	25.9	53.2	24.6	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.4	(b) 10.4	(c) 9.3	(e) -35.8	(f) 34.9	(g) 2.0	(h) 2.6	(i) -37.2	(j) 36.8	(k) -38.4	(l) 38.3	(m) -39.5	(n) 39.7	(o) -40.9	(p) 41.6		
(4) 12.4	(5) 10.4	(6) 9.3	(7) -36.0	(8) 21.1	(9) 2.0	(10) 1.2	(11) -37.4	(12) 22.0	(13) -38.6	(14) 22.7	(15) -39.7	(16) 23.3	(17) -41.1	(18) 24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.9	-21.3	-16.9	-7.3	3.8	11.2	17.5	20.3	18.0	11.5	1.8	-10.6	-18.9	(6)
(7)		DBStd	15.53	5.01	5.77	6.86	5.00	4.76	4.12	3.19	3.61	4.70	5.15	7.02	5.28	(7)
(8)		HDD10.0	4302	971	754	536	193	42	1	0	1	36	257	617	895	(8)
(9)		HDD18.3	6541	1229	987	794	435	226	63	14	49	211	512	867	1153	(9)
(10)		CDD10.0	967	0	0	0	8	80	226	321	249	80	3	0	0	(10)
(11)		CDD18.3	164	0	0	0	0	5	38	76	40	5	0	0	0	(11)
(12)		CDH23.3	2062	0	0	0	6	131	559	814	465	87	0	0	0	(12)
(13)		CDH26.7	718	0	0	0	1	34	204	307	153	18	0	0	0	(13)
(14)	Wind	WSAvg	3.8	4.1	3.8	3.9	4.7	4.8	3.6	3.0	2.9	3.6	3.7	3.6	4.1	(14)
(15)	Precipitation	PrecAvg	283	2	2	3	11	16	46	80	74	30	10	6	3	(15)
(16)		PrecMax	448	6	7	17	39	34	111	176	168	96	28	23	11	(16)
(17)		PrecMin	188	0	0	0	0	1	1	29	18	2	0	0	0	(17)
(18)		PrecStd	56	2	2	4	11	10	27	35	36	24	7	6	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.0	2.0	14.4	23.4	30.3	34.5	35.9	33.3	29.3	20.2	8.2	-3.8	(19)
(20)			MCWB	-7.7	-2.0	4.5	9.4	13.2	16.7	19.2	18.4	14.9	9.6	2.1	-5.7	(20)
(21)		2%	DB	-9.4	-2.0	9.7	19.1	26.7	31.4	32.6	30.4	25.8	16.2	4.9	-6.9	(21)
(22)			MCWB	-10.7	-4.7	2.4	8.2	12.7	15.7	18.2	17.2	14.1	7.4	0.3	-8.3	(22)
(23)		5%	DB	-11.7	-5.1	6.0	16.2	24.0	28.9	30.0	28.1	23.1	13.2	2.0	-9.3	(23)
(24)			MCWB	-12.7	-7.0	0.4	6.6	11.6	15.3	17.7	16.5	12.7	5.7	-1.6	-10.4	(24)
(25)		10%	DB	-13.9	-7.6	3.1	13.5	21.4	26.5	27.7	25.7	20.5	10.6	-0.6	-11.2	(25)
(26)			MCWB	-14.7	-9.0	-1.3	5.1	10.3	14.6	17.1	16.0	11.8	4.2	-3.3	-12.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.6	-1.8	4.9	10.4	14.9	18.9	20.7	20.4	16.7	9.9	2.4	-5.5	(27)
(28)			MCDWB	-6.2	1.5	12.7	20.4	27.0	29.1	30.1	29.6	25.4	19.8	7.9	-3.8	(28)
(29)		2%	WB	-10.7	-4.7	2.5	8.5	13.4	17.4	19.7	19.0	15.0	7.6	0.4	-8.2	(29)
(30)			MCDWB	-9.5	-2.2	9.1	18.5	24.5	27.0	28.6	25.9	22.0	15.1	4.8	-6.9	(30)
(31)		5%	WB	-12.7	-6.9	0.5	6.8	12.0	16.3	18.9	18.0	13.8	6.0	-1.4	-10.4	(31)
(32)			MCDWB	-11.8	-5.1	5.7	15.7	22.4	25.8	27.1	24.4	20.5	12.6	1.8	-9.3	(32)
(33)		10%	WB	-14.7	-9.0	-1.2	5.2	10.7	15.4	18.0	16.9	12.6	4.5	-3.2	-12.1	(33)
(34)			MCDWB	-13.9	-7.6	2.9	13.1	20.5	24.4	25.4	23.4	19.4	10.1	-0.8	-11.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	12.7	13.5	14.0	15.3	13.7	12.2	12.8	14.1	13.1	11.4	9.3	(35)
(36)			MCDBR	11.7	15.1	16.6	18.6	19.6	17.8	15.9	16.8	17.7	17.1	14.2	10.5	(36)
(37)			MCWBR	10.6	12.7	10.9	10.3	9.1	6.8	5.6	6.6	8.5	10.0	10.2	9.4	(37)
(38)		5% WB	MCDWB	11.6	14.7	15.7	17.4	17.3	14.6	13.3	13.0	15.0	16.2	13.8	10.3	(38)
(39)			MCWBR	10.7	12.6	10.5	10.0	8.6	6.3	5.6	6.0	7.6	9.8	9.9	9.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.248	0.272	0.311	0.373	0.431	0.436	0.405	0.372	0.332	0.297	0.265	0.250	(40)	
(41)		taud	2.323	2.286	2.268	2.156	2.027	2.131	2.289	2.355	2.404	2.427	2.385	2.304	(41)	
(42)		Ebn at Noon	846	895	906	875	833	830	853	869	876	857	816	790	(42)	
(43)		Edh at Noon	70	92	111	139	165	149	126	113	97	80	66	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.95	3.12	4.60	5.72	6.50	6.44	6.14	5.62	4.57	3.20	2.04	1.54	(44)	
(45)		RadStd	0.12	0.18	0.18	0.23	0.30	0.43	0.29	0.31	0.29	0.18	0.10	0.12	(45)	

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(46) Station Only	DBAvg	N/A	N/A	-1.17	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	DBAvg	N/A	N/A	-1.51	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page